Investments
Autumn 2026

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Homework #3 The Capital Asset Pricing Model

This problem set is due Week 4. You are to submit solutions to the assignment using Canvas. You may work on this in groups of your choosing. Groups may be of no more than 5 people. Submit one solution for each group following the instructions on Canvas.

PLEASE NOTE: Many of the questions ask you to report results from data summary and/or estimation. Please do not just provide raw output from Excel, Matlab or any other software. You are to provide answers in the form of neatly formatted tables or labeled graphs.

1. You are given the following information: the variance of the rate of return on stock 1, stock 2, and the market portfolio are, $\sigma_1^2 = 0.16$, $\sigma_2^2 = 0.09$, and $\sigma_M^2 = 0.04$. The covariances between these assets are $\sigma_{1,2} = 0.02$, $\sigma_{1,M} = 0.064$, and $\sigma_{2,M} = 0.032$. Consider forming a portfolio that has 75% invested in asset 1 and 25% invested in asset 2. Call this portfolio p.

   (a) What is the variance of portfolio p?

   (b) What are the betas of 1, 2, and p relative to the market (i.e., what are $\beta_{1,M}$, $\beta_{2,M}$, and $\beta_{p,M}$)?

   (c) What are the R^2 values in regressions of the return on 1, 2, and p on the market portfolio?

2. Obtain the data file **ps3_problem2.xlsx** from Canvas. This file contains historical monthly rates of return on U.S. and global stock indices, treasury bills, and individual stocks. The following calculations can be performed in Excel.

   (a) Set up four different portfolios, finding monthly rates of return on each. The first portfolio is evenly invested in GM and IBM.

The second portfolio is evenly invested in three stocks (GM, IBM, and Anheuser Busch) while the third portfolio is evenly invested in four stocks (GM, Toyota, IBM, and Anheuser Busch). The last portfolio is equally invested in all 13 common stocks. The rate of return on any portfolio is just the weighted-average of the rates of return on the underlying stocks.

i. Calculate the average monthly return and the return variance on these four portfolios (you do not need to use any complicated formulas here, just apply Excel functions AVERAGE and VAR to the portfolio returns).

ii. For each of these four portfolios, *estimate* the beta (β) and R^2 of the portfolio versus the "World Market" portfolio. (If you are working in Excel, the functions SLOPE and LINEST will both estimate the beta. LINEST provides other regression statistics as well.)

(b) How does the variance of the portfolio change as we add assets to the portfolio? How does the beta change as we add assets to the portfolio?

(c) What do the results in part (2a) tell us about the behavior of the components of risk and return as we add assets to the portfolio?